

Semester Project: Smart Parking Platform

Project Overview

WE ARE \$oftware \$orp. A fictitious software development studio with unlimited resources and budget.

The project involves the planning and management of a software system that helps drivers locate, reserve, and pay for parking spaces for parking garages in real time. The platform will be available through web and mobile applications and will provide parking availability information, navigation assistance, and management tools for parking operators.

The goal is to develop a project management plan for the creation of this system, including requirements gathering, stakeholder management, scheduling, budgeting, risk management, and team coordination.

Business Need

Drivers often spend significant time searching for available parking, leading to congestion, frustration, and lost productivity. Parking operators need better tools to monitor occupancy, manage pricing, and improve utilization of parking assets.

High-Level Requirements

The platform should provide:

- User registration and authentication
- Real-time parking space availability
- Interactive map displaying available parking locations
- Parking reservation capability
- Digital payment processing
- Reservation history and receipts
- Administrative dashboard for parking operators
- Occupancy reporting and analytics
- Notifications and alerts for users
- Integration with external mapping/navigation services

Additional requirements should be identified through stakeholder analysis and requirements elicitation activities.

Stakeholders

- Drivers and vehicle owners
- Parking facility operators
- City transportation departments
- System administrators
- Finance and billing personnel
- Mobile application users
- Third-party payment providers

Development Teams

- Mobile Application Team
- Web Application Team
- Backend/API Team
- Mapping & Location Services Team
- Payment Integration Team
- Quality Assurance & Testing Team

Project Constraints

- Unlimited theoretical development budget
- Fixed semester timeline
- Dependence on third-party services and APIs
- Compliance with security and privacy requirements, city ordinance & laws

Research Component

You must perform a competitive analysis of existing parking solutions and identify strengths, weaknesses, and opportunities for differentiation.

Weekly updates to the project document

Every week's homework will start a part or section of the final project document. The weekly requirements are not 100% complete, and it is every student's responsibility to complete each

component until the final project is due at the end of the semester.

Additionally, ensure that the final project document is well formatted, includes a header and a footer, a table of contents, and image captions.